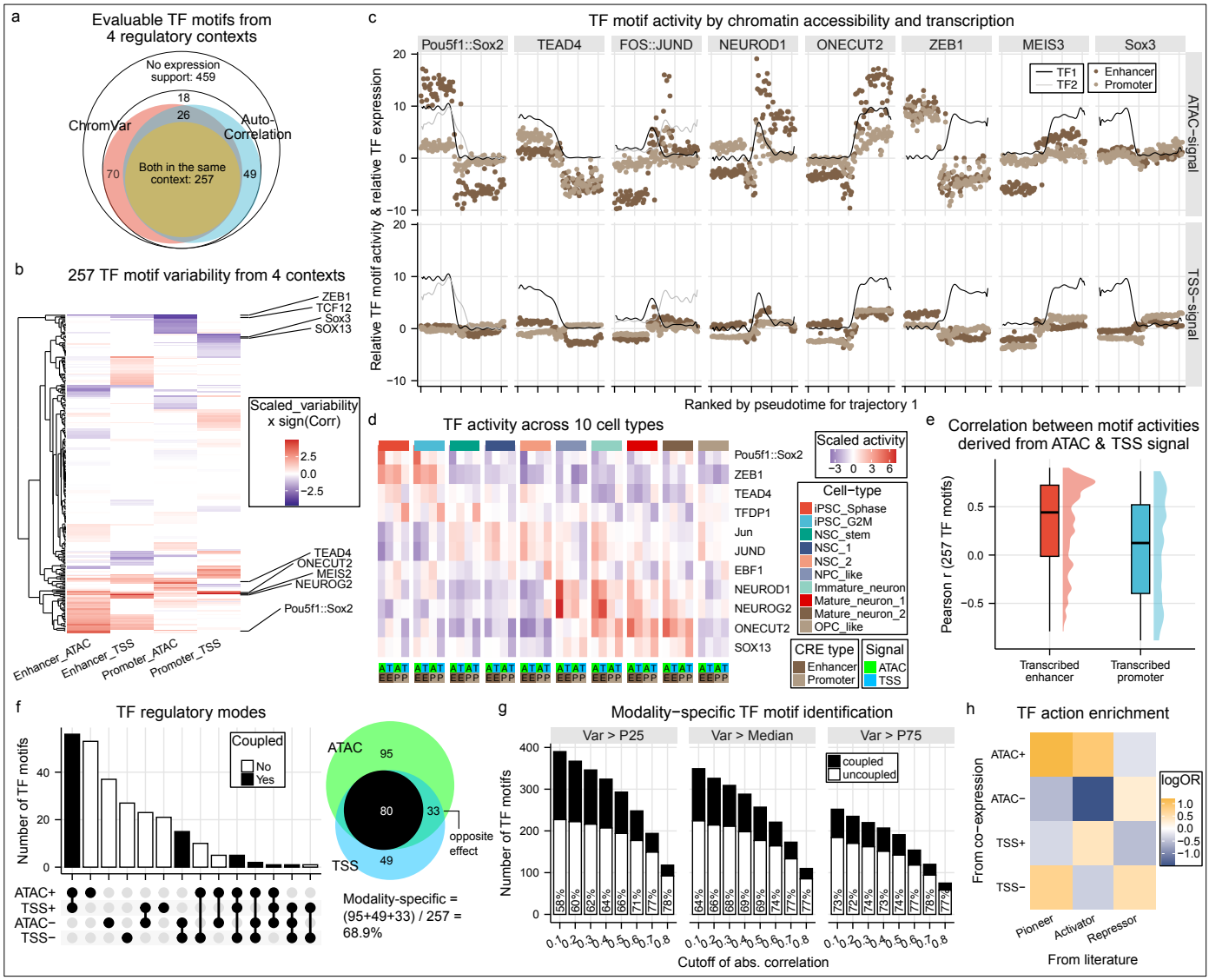


Figure 3 | TF regulatory modes reveled from dynamic TF expression and motif activity.

a, Schematic summarizing the filtering of evaluable TF motifs from this experimental platform. From the initial database (879 motifs from Jasper 2024), 420 TF motifs were selected based on TF expression (normalized expression > 0.2 from at least one single cell across at least 3 metacells, applied to both TFs for dimeric motifs). From these, 257 TF motifs showed significant and strong variability and correlated with corresponding TF expression from the same conditions. **b**, Heatmap showing TF motif variability of the 257 evaluable motifs across four regulatory contexts. Scaled variability is colored by the sign of correlation, where positive values indicate activity supported by TF and negative values indicate activity suppressed by TF. **c**, Relative TF motif activity derived from ATAC (upper) and TSS (lower) signals and relative expression level of TFs were shown across differentiation pseudotime. The z-scores derived from ChromVar for each condition were utilized as relative motif activity. **d**, Heatmap showing summarized motif activity across the 10 cell clusters with four regulatory contexts from 12 selected TF motifs. Values represent the average scaled ChromVar z-scores per clusters, illustrating dynamic regulatory transitions. **e**, Pearson correlation of motif activity scores (chromaVar z-score for each metacell) derived from ATAC-signal and TSS-signal from enhancers and promoters. The correlation coefficient from 257 TF motifs are shown. **f**, Left: upset plot summarizing the number of TF motifs categorized by their regulatory modes. Right: venn diagram showing the overlap of 257 evaluable TF motifs using ATAC and TSS signals. **g**, Percentage of TF motifs that are modality-specific according to the definition from (**f**) with different combinations of cutoff from ChromVar variability and correlation. **h**, Odd ratios from the enrichment between different modes of action identified in this study and established literature across 184 TF motifs with information from both ends.